

Week 9 Bias-variance

1. What is regularization?

regularization is a technique used to prevent overfitting in a predictive model. Overfitting occurs when a model is too complex and captures noise and randomness in the training data, which results in poor performance on new, unseen data.

Regularization introduces additional constraints or penalties to the model during training, which encourages it to prefer simpler models that are less likely to overfit. These penalties can take different forms, but they all aim to reduce the model's flexibility by discouraging large parameter values or complex relationships between features.

Two common types of regularization are L1 and L2 regularization. L1 regularization, also known as Lasso regularization, adds a penalty to the absolute values of the model's parameters, which encourages sparsity and feature selection. L2 regularization, also known as Ridge regularization, adds a penalty to the squared values of the model's parameters, which encourages small parameter values and reduces the impact of outliers.

- **Replace**

- $-Loss(X, t, w)$ with
- $-Loss(X, t, w) - \alpha R(w)$

When training towards the original learning objective, there is a danger for overfitting, which means that the trained model fits the training set very well, but it does not generalize equally well to other data. In particular, some features might be particularly important and by repeated applications of gradient descent, the model ascribes more weight to these features. **The goal of regularization is to avoid putting too much weight on some features, and instead try to get more even weights between the features.** b) Alpha determines how much larger weights get punished. There is no fixed α which fits all cases. The optimal α depends on the task in question. α is tuned experimentally. One separates the development data into a training set and a development test set. For various values of α , one trains models on the training set and evaluates on the development test set. The value of α which gives the best score on the development test set is chosen for the final model.

2. What is bias-variance tradeoff?

Bias-variance tradeoff is a fundamental concept in machine learning that describes the relationship between a model's bias and variance, and their impact on the model's performance.

Bias refers to the difference between the expected predictions of a model and the true values. A high bias model is too simplistic and may underfit the data, resulting in poor accuracy on both the training and test sets. **Variance refers to the variability of the model's predictions across different samples of the training data.** A high variance model is too complex and may overfit the data, resulting in excellent performance on the training set but poor performance on the test set.

The bias-variance tradeoff suggests that as a model becomes more complex, its variance increases and its bias decreases. Conversely, as a model becomes simpler, its variance decreases and its bias increases. The goal is to find the right balance between bias and variance that results in the best performance on new, unseen data.

Regularization techniques, such as L1 and L2 regularization, can be used to reduce variance and increase bias by adding constraints to the model's parameters. In addition, techniques such as cross-validation can be used to estimate a model's bias and variance and find the optimal tradeoff for a given problem.

3. What is cross-validation?

Cross-validation is a technique used in machine learning to evaluate the performance and generalization ability of a model. It involves dividing a dataset into two or more subsets: a training set, a validation set, and sometimes a test set.

The training set is used to train the model, while the validation set is used to tune the hyperparameters of the model and select the best performing model. The test set is used to estimate the model's performance on new, unseen data.

Cross-validation involves repeating this process multiple times with different partitions of the data to obtain a more robust estimate of the model's performance. One common form of cross-validation is k-fold cross-validation, where the data is divided into k subsets of equal size. The model is trained k times, each time using a different subset as the validation set and the remaining subsets as the training set. The performance metrics are then averaged across the k runs to obtain a more reliable estimate of the model's performance.

Cross-validation is important because it helps to avoid overfitting

In machine learning, one applies algorithms that improve automatically through experience. The algorithms are trained by the help of data, a training set. To see that the algorithms have learned, one needs a precise description of what the learning objective is, and ways to measure improvement with respect to this objective. To measure the improvement and how well the system performs, one uses a test set of relevant data. The test set must be disjoint from the training set. Often, one would compare different ML systems trained on the same data. For this, one needs a development test set different from both the training set and the test set. One may train repeatedly different systems on the same training set and test them on the development test set. One then decides on the system which performs best on the development test set. After deciding on the best system, this can be tested on the (final) test set, which hasn't been considered so far.

4. What is bagging and random forests?

Bagging (Bootstrap Aggregating) and Random Forests are ensemble learning techniques used in machine learning to improve the accuracy and robustness of predictive models.

Bagging involves generating multiple training sets by randomly sampling the original dataset with replacement. Each training set is used to train a separate model, and the final predictions are obtained by averaging the predictions of all the models. Bagging helps to reduce the variance of the predictions and improve the model's stability.

Random Forests are an extension of Bagging that builds a large number of decision trees, each trained on a subset of the features and a subset of the data. In addition, at each node in each tree, a random subset of the features is considered for the split, rather than considering all the features. The final predictions are obtained by averaging the predictions of all the trees. Random Forests help to reduce the correlation between the trees and improve the model's performance.